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Predicting Outcome of 2020 Presidential Election

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DATASCI 415
https://github.com/tjgirdler/415-Final-Report

1 Introduction

With the US presidential election happening this year, our group was interested in learning if election results can be predicted using data science skills. Since the 2024 election had not happened at the start of the project, we decided to see how well we could predict the 2020 election results by county. To try and predict the winner for each county, we gathered demographic data about the counties from the US Census Bureau. We also gathered the results of every presidential election by county from 2000-2020 from the MIT Election Data and Science Lab. With these data, our goal was to build a model that could predict the winner of the 2020 presidential election in each county with 95 percent accuracy.

Since all presidential elections since 2000 have been a close race between Democrat and Republican candidates, our target is a binary categorical variable called "winner-2020". This target has a value of 0 for counties that had a majority vote for Donald Trump in 2020, and a value of 1 for counties that had a majority vote for Joe Biden. Some simple linear regression models were also built that tried to predict the margin of votes between parties in 2020. The full dataset had 37 continuous features which included various demographics of county populations and the margin of votes between the democratic and republican candidates of past presidential elections. We built prediction models using linear regression, simple classification methods, support vector machines, and decision trees.

2 Related Work

There are many models people have built to try to predict election results using various machine learning methods. We have included some works related to different research questions that use techniques similar to those used in this project.

There are works that use various simple classification techniques on similar problems, specifically clustering. One example is a method by Schmitz et al.[3] based on South African elections and it is applied to US elections. They used past election results to create statistical clusters, using the Euclidean distance as their similarity measure, to predict how a county would vote. This approach was most similar to our KNN method which also found that grouping counties based on their previous voting records was very indicative of their 2020 vote. A different study done by Saif Gazali and V. Pattabiraman[6] included polling data, twitter sentiment analysis, as well as previous election results as features for their model. Using logistic regression and Naive Bayes, they obtained an accuracy of 86% both times. We found this to be a good goal for our simple methods.

Related works researching decision trees stressed the importance of pruning trees to reduce the number of predictors. Marzuki et al.[2] looked at predicting the outcome of an individual voter, obtaining a test accuracy of 84.27% using a heavily pruned model with only 3 decision rules and 5 nodes. Santiago Olivella Jacob M. Montgomery.[8] highlighted the utility of trees for handling interactions in nonlinear relationships, and mentioned that tree ensembles are particularly good for

[1] https://github.com/tjgirdler/415-Final-Report

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

these cases. Anthony Scime Gregg R. Murray[7] focused on targeting factors for individual voters. Due to a far more specified problem, the prediction accuracy for different segments of voters ranged from 66% to 87%.

Pankaj Sinha et al.[15][14] proposed and tested regression models to predict the vote share of the incumbent and non-incumbent parties in the 2016 and 2020 presidential elections for the Munich Personal RePEc Archive. They used a combination of economic and non-economic features to build a simple linear regression model. Pankaj Sinha et al.[16] and D Irvani et al.[1] used Lasso as a shrinkage method for their machine learning models predicting election results, using cross-validation for selecting the penalty term and RMSE as a measure of performance. Due to our concern for multicollinearity, we opted to use ridge regression as a shrinkage method, similarly to Sanatan Mishra and John Lee[10], who built a model to predict US House of Representative election results.

Chitra Jegan[9] built a Support Vecrtor Machine Classifier in an attempt to predict whether or not a person had diabetes using a radial kernel and 10-fold cross validation. In our SVM, we similarly used a radial kernel and 5-fold cross validation. Amit Choudhary and Savita Ahlawat[12] used a hybrid CNN and SVM approach to build a classification model using a radial kernel.

3 Dataset and features

Our dataset is a combination of three data sets: demographic information from counties, presidential election results from 2000 to 2020, and separate presidential election results for Alaska. The election results from 2000-2016 were coded as the proportional difference of people who voted democrat and the people who voted republican. The rows in our data set represent counties in the United States. There is voting information from previous elections, as well as information of the makeup of counties. For example, some features include median household income and proportion of Hispanic/Latino residents. Counties lacking voting information were dropped from the data set, leaving us with 3,136 counties out of 3,144 total counties.

Once most of the dataset was finalized, special considerations had to be taken for Alaska. Alaska does not split up into counties, rather boroughs and planning regions. They also do not release voting information the same way that every other county does. This is why we had to separately gather estimated voting data for Alaska for elections since 2000 for 30 boroughs. This information was joined with the demographic dataset, which did have economic data, race data, etc. based on Alaskan boroughs.

We used a 20% held-out test set in each model.

4 Methods

We used a variety of methods and feature subsets to try and predict 2020 election results with 95% accuracy.

4.1 Using all available features

First, we used simple methods to get an understanding of the dataset. We began with simple classification methods that predicted whether each county had a majority vote for "Democrat." Before beginning, the variable 'State' was one-hot encoded, adding 49 additional features to the data set. We used simple methods such as logistic regression, LDA, QDA, Naive Bayes, and KNN. They were all also run without the encoded 'State' variable, which ultimately did not make a big difference. LDA, being the standout model, is explained by finding a linear decision boundary to separate the data that minimizes the spread of data in each class. It is described using the equation:

$$\delta_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k)$$

Where a sample (county) is classified into the class with the highest discriminant score. Priors (π_k) are calculated using the training set. A linear boundary is created using $\mathbf{w}^T \mathbf{x} + b = 0$ where $\mathbf{w} = \Sigma^{-1}(\mu_1 - \mu_2)$.

Finally, we used model selection to find the best variables. This was done using logistic regression, with Mallow's Cp as the stepwise selection method. We ran the same process using LDA as the model, because this model performed best out of all the simple methods. Both times, the model selection only favored two variables: the proportion of people who voted democrat in 2012 and 2016.

In addition to simple classification, we also made classification trees to classify counties. The method used to determine splits in the tree was cross-entropy. $D = -\sum_{k=1}^{K} \hat{p}_{mk} log(\hat{p}_{mk})$. The tree with all 86 predictors was also pruned using cross-validation. Using both of these methods, it was clear that previous voting records, especially those of the last two election cycles, were the most useful in the model. We decided to see how well the models could perform using only these two variables.

4.2 Using only past election results

The LDA model was run again with just the two variables (proportion each county voted democrat in 2012 and 2016). After this initial exploration, some more work was done using the KNN method, as this was most similar to the clustering methods used in the field. This method used the Euclidean distance and predicted the average of the K most similar counties found. Written out, the distance found between each pair of points is:

$$d(\mathbf{x}, \mathbf{x}_i) = \sqrt{\sum_{j=1}^{p} (x_j - x_{ij})^2}$$

A prediction is done for each point using the response mode (0 or 1) of the nearest K points. The KNN model also performed best with only the aforementioned predictors, along with standardization and cross-validation to choose the number of neighbors (around 20).

We also ran OLS with the two variables used in the previous LDA model to predict the proportion each county voted democrat in 2020. OLS will estimate the regression coefficents of the two variables (β_1 and β_2) by minimizing the RSS, denoted by $RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$. In this case, $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}$ and y_i are the true democrat proportions for 2020. Since we are only considering two variables in this case, n=2. Once we obtained estimates for β_1 and β_2, we were able to compute the MSE between the training and test proportions to assess model performance, which came out to be very small.

Since these two predictors dominated the performance of the model, after simple classification was done, we wanted to see if we could find other useful predictors. We began to run models *without* the results of the previous elections to find out if other variables could be strong predictors.

4.3 Without past election results

To gain a stronger understanding of the factors that are associated with voting behavior, we will omit past election results from the models and remove variables that exhibit multicollinearity in this section, resulting in models that begin with 17 variables.

Beginning with OLS, we considered backward selection, non-linearity, and interactions to improve the model performance. Backward selection attempts to find the best subset of predictors that are all statistically significant towards the model, which is useful for model interpretation. From the backward selection of variables, we implemented interaction terms and non-linearity to the model with the desire to improve model accuracy by reducing the MSE.

We also implemented Ridge Regression to potentially reduce our MSE further. Ridge Regression adds a penalty term to the RSS, which serves as a way to constrain the regression coefficients of OLS models. The Ridge Regression coefficient estimates β will minimize:

$$\sum_{i=1}^{n} (y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})^2 + \lambda \sum_{j=1}^{p} \beta_j^2 = RSS + \lambda \sum_{j=1}^{p} \beta_j^2$$

where the hyperparameter λ is selected by cross-validation to ultimately produce the lowest MSE in testing. Given that Ridge Regression is able to account for multicollinearity, we considered some of

the variables that were previously removed. The Ridge Regression model used 25 variables instead of 17.

These methods will provide meaningful results towards predicting the margin of votes between parties, which can be used to predict the winner of an election, but classification methods may serve as a more direct way to address our primary interest.

Next, we built a support vector machine classifier. Support vector machines attack classification problems by find a hyperplane that can best separate the two classes, in this case, county winners Donald Trump and Joe Biden, coded as -1 and 1 for the SVM. The equation of the hyperplane takes the form:

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_p X_p = 0$$

where p is the number of features.

The SVM uses a Maximal Margin Classifier (denoted M) which is design to maximize the βs and ϵ over M subject to $\sum_{j=1}^{p} \beta_j^2 = 1$, $y_i(\beta_0 + \beta_1 x_i 1 + ... + \beta_p x_i p) \geq M(1 - \epsilon_i)$ for all $i = 1, ..., N$, $\epsilon_i > 0$, and $\sum_{i=1}^{n} \epsilon_i \leq C$, where C is the tuning parameter.

Due to issues with multicollinearity and overfitting, all but 17 features were removed from the model and the remaining were standardized. Using test accuracy as the measure of a good model, we built an initial classifier with all variables, using a linear kernel and a tuning parameter of C=10, giving us a test accuracy of 0.93. With the goal of beating this accuracy score, we looped through linear and radial kernels, along with possible values for the hyperparameters to find the best model. The hyperparameters we tried included:

'linear': 'C': [0.001, 0.01, 0.1, 1, 5, 10, 100]

'rbf': 'C': [0.001, 0.1, 1, 5, 10, 100], 'gamma': [0.0001, 0.001, 0.01, 0.1, 1]

In addition to running classification trees on all variables, we also ran them without past election results. Since collinearity is a problem for decision trees, we chose to make them using the same features as the SVM models. We made pruned trees and random forests. We did this on the data without election results and then made trees with and without states as predictors as well.

5 Experiments/Results

Features Used	Model	Accuracy or MSE
All Features	LDA	97.13%
All Features	Decision Tree	97%
Only Previous Elections	KNN	98.01%
Only Previous Elections	OLS	0.0024
Without Previous Elections	Ridge	0.0344
Without Previous Elections	SVM	93.30%
Without Previous Elections	Random Forest	93.31%

5.1 Using all available features

The most successful simple classification method before any model selection was LDA. These are the results without the 'State' variable, however. The LDA test confusion matrix shows that the model does a good job. The overall test accuracy was 97.13%. It is more likely for the model to predict Republican over Democrat, even when wrong, which may be due to class imbalance.

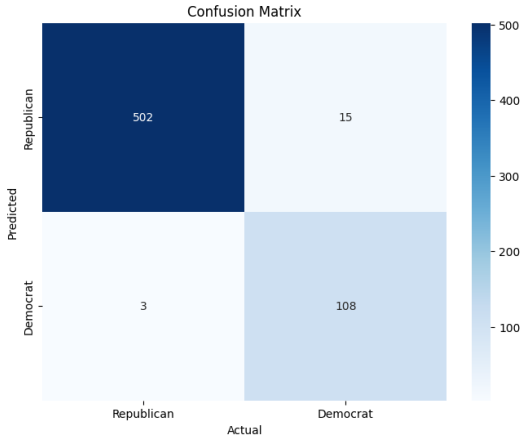


Figure 1: LDA Full Model Confusion Matrix

The decision tree, including former election voting proportions with all other predictors from the dataset, had a 97% test accuracy. It was 91% accurate on Democratic counties and 99% accurate on Republican counties. Additionally it produced a tree with only 5 rules and 6 leaves, solely determined by the proportion the Democratic Party won by in 2016, median household income, and proportion of Hispanic people in a given county. To view the decision tree, see Appendix 8.

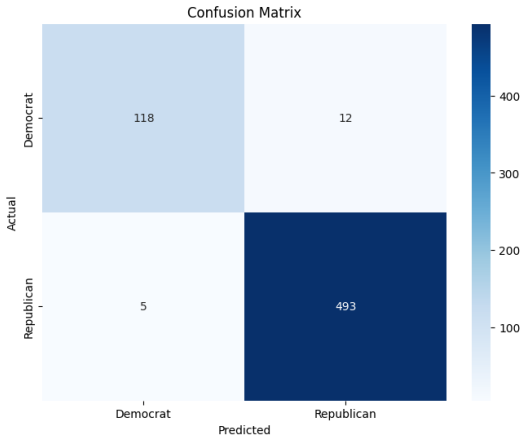


Figure 2: Confusion Matrix for the Decision Tree

5.2 Using only past election results

After model selection was applied, only two predictors were found to be useful when applying LDA. These were plotted with the LDA decision boundary, using the full dataset. This accuracy was also not that different from the LDA model without predictor selection. As seen, just using these two predictors, a fairly good LDA decision boundary can be drawn. This shows the potency of just these two predictors, and gives us an interpretable result of election results. There are hardly any outliers, with only one county very clearly predicted majority democrat despite being majority republican.

Train Accuracy	97.64%
Test Accuracy	97.45%

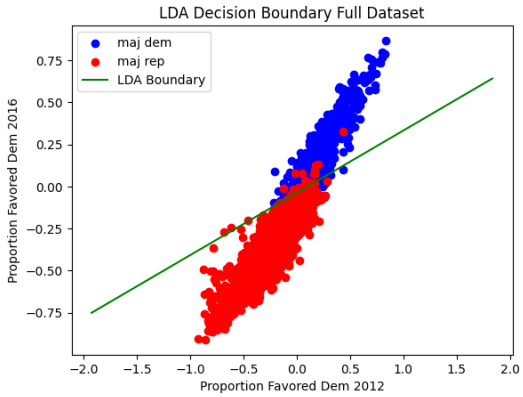


Figure 3: LDA Full Dataset

The most successful KNN model, which included a cross-validated K of 15, standardizing, and only using the two previous election results as predictors, was also very successful. This performed slightly better than the LDA model(s), but not notably different. The test accuracy was 98.01%.

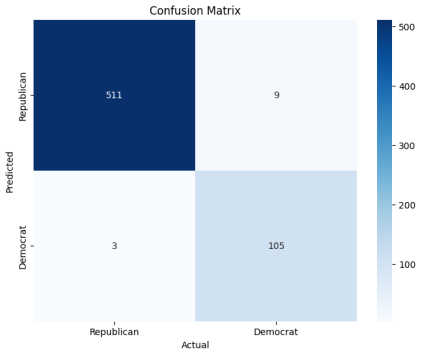


Figure 4: Final KNN Model Confusion Matrix

These same two predictors used in OLS returned similar high performance. The MSE between the predicted and actual 2020 democrat proportion was 0.0024. In consideration of our problem at hand, we classified the predicted and actual 2020 democrat proportions by determining whether the proportion was positive or negative. Positive proportions correspond to a democratic majority, while negative proportions correspond to a republican majority. Using only the two predictors, this classification returned an accuracy of 97.45%. The accuracy of these results is not surprising given that the proportion of Democratic votes in previous elections tend to be very deterministic of the proportion for the following election. Voting trends for a county rarely change drastically from election to election. Although we obtained a very low MSE, it does not give us much information in terms of the predictors that are important in predicting election results other than the two "obvious" ones.

5.3 Without past election results

To gain an understanding of other predictors that are useful in predicting election results, we created an OLS model initially with the 17 predictors. Using only backward selection at a significance level of 0.05, the model was reduced to 10 predictors that considered factors such as age, education levels, and ethnicity/language. Using the same classification conversion described in the previous section, the model produced an accuracy of 89.81% and an MSE of 0.0359. Once we included interactions and non-linear terms, the accuracy increased to 90.13% and the MSE decreased to 0.0344. Interacting the two education variables (Bachelor's Degree or Higher and High School or Higher percentages) allowed the model to capture the relationship that those who complete high school often move onto

college. This interaction along with the non-linear fittings of other variables slightly improved our OLS model.

We ran Ridge Regression with 25 predictors to allow the model to account for the multicollinearity between variables that were previously removed. The λ selected using cross-validation was just over 0.01, which led us to estimate the MSE of Ridge Regression to be 0.0344. While this model returned the same MSE as the final OLS model, the Ridge Regression did not consider any interactions or non-linear terms. Compared to the OLS model with only backward selection, Ridge Regression was able to slightly reduce the MSE. We were able to improve the prediction performance obtained from linear regression by implementing classification methods.

One of the most successful prediction model for 2020 election results that did not use past election results was a support vector machine classifier with a radial kernel. Using 5-fold cross validation, the model's hyperparameters were tuned as follows:

Kernel	'rbf'
Tuning Parameter	C = 100
Gamma Parameter	gamma = 0.01
Test Accuracy	93.30%

The following ROC curve shows the tradeoff between the true positive and false positive prediction rates. The an Area Under Curve (AUC) of 0.97 tells us that our model predicted election results relatively well. The confusion matrix shows the accuracy of predictions for Democratic and Republican counties.

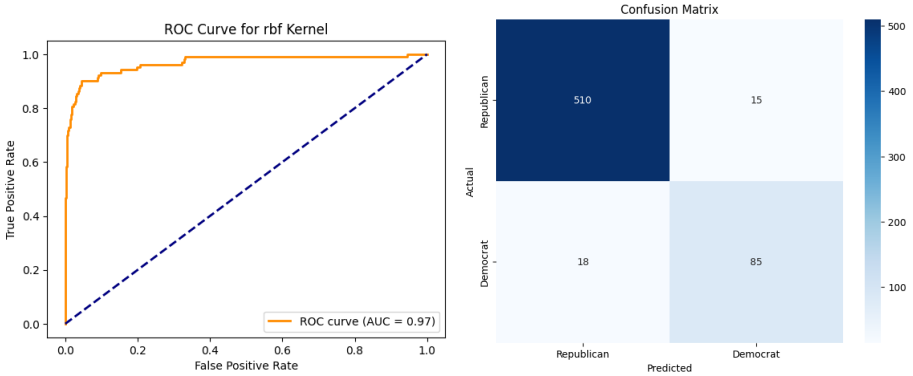


Figure 5: ROC Curve and Confusion Matrix for Radial Kernel SVM Classifier

In addition to regression and SVM's, we also ran decision trees without past election results. The state predictor proved to be fairly ineffective as trees without states involved performed similarly to trees with them.

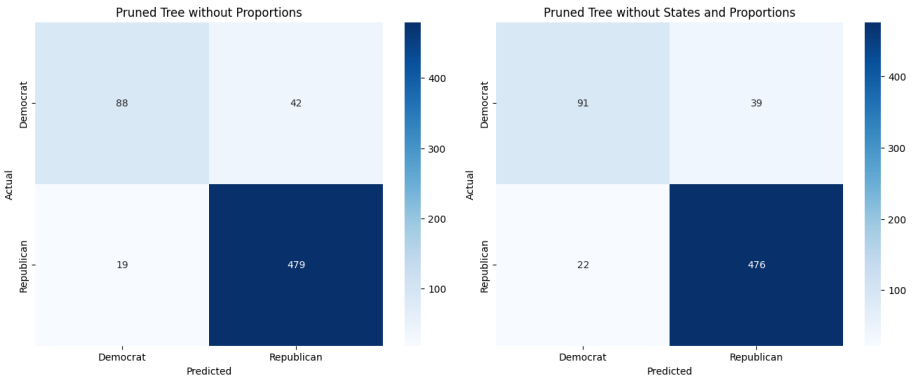


Figure 6: Confusion Matrices for Pruned Trees

Random forests offered a significant improvement over pruned trees, going from 90.29% accurate in pruned trees to 93.31% accurate in the random forest with states and 93.15% accurate in the random forest without states.

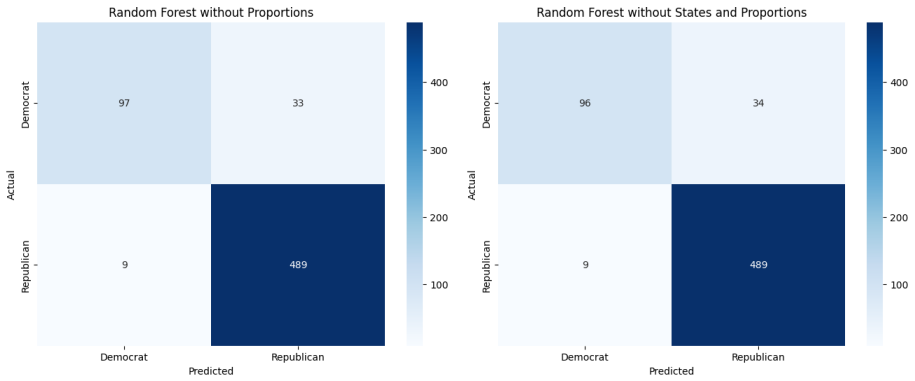


Figure 7: Confusion Matrices for Random Forests

6 Conclusion/Discussion

6.1 Conclusion

When including the results of past elections, our best prediction model for the 2020 presidential election was an LDA model that obtained a test accuracy of 97.13%. When those features were removed, both the radial support vector machine and the random forest that included states had the highest accuracy of about 93.3%. When we used only the past election results as predictors, the best model was a KNN classification model with a 98% test accuracy.

We learned that the most important factors in predicting the election results were very clearly past election data. This makes intuitive sense because in general, the makeup of a given county would not change much over the course of 8 years. Another takeaway was that a large majority of counties (>82%) were Republican in the full dataset. Due to this, many models were successful by predicting Republican majority.

We wanted to learn more than just predicting based on past election results, and as seen during our methods, we saw what other factors were important after removing these predictors. Based on the decision tree models, some of the most important predictors for counties included the percentage of non-Hispanic or Latino white people, percent of people with a Bachelor's degree or higher, population per square mile, per capita income, and foreign variables (Foreign Born, Language Other than English at Home). Using the OLS regression coefficients without interactions or non-linearity, we assessed whether these predictors were associated with more democratic or republican voters. Democratic counties tended to achieve higher educational levels, had larger population densities, obtained higher per capita income, and had more non-English speaking households. Republican counties tended to have more white people and more foreign born residents.

6.2 Limitations

Some limitations of our project was that our data did have a large class imbalance, impacting models like the Logistic model, which always predicted 0. Another limitation was that the demographic data was measured only for 2020, so we could not compare these with demographics and election results of previous years. If we were to re-do this project, we might try and find demographic data for past election years so we could use these results as training data, rather than predictors.

Furthermore, additional features containing information about other political figures that were elected by the counties might paint a better picture of where these areas stand politically. Although we did make an attempt to add geographical data (longitude and latitude) [13] it was not useful in the model and was difficult to perfectly match with counties that are constantly changing. This may also be an interesting method to further explore in the future.

7 Contributions

Each team member handled a set of models to predict the 2020 election, and found related articles, investigated methods, and gathered results regarding them. The introduction, conclusion, and limitations were written as a group.

7.1 Padmalekha Danturty

- Preprocessing Alaska voting data
- Simple Classification Techniques (Naive Bayes, Logistic Regression, QDA, LDA, KNN)

7.2 Maisie Fujita

- Support Vector Machine Classifiers
- Wrote Related Works section

7.3 Thomas Girdler

- Regression Methods (OLS, Ridge)
- Creating References section

7.4 Malachi Mealoy

- Merging Final Data set
- Decision Tree Models

8 Appendix

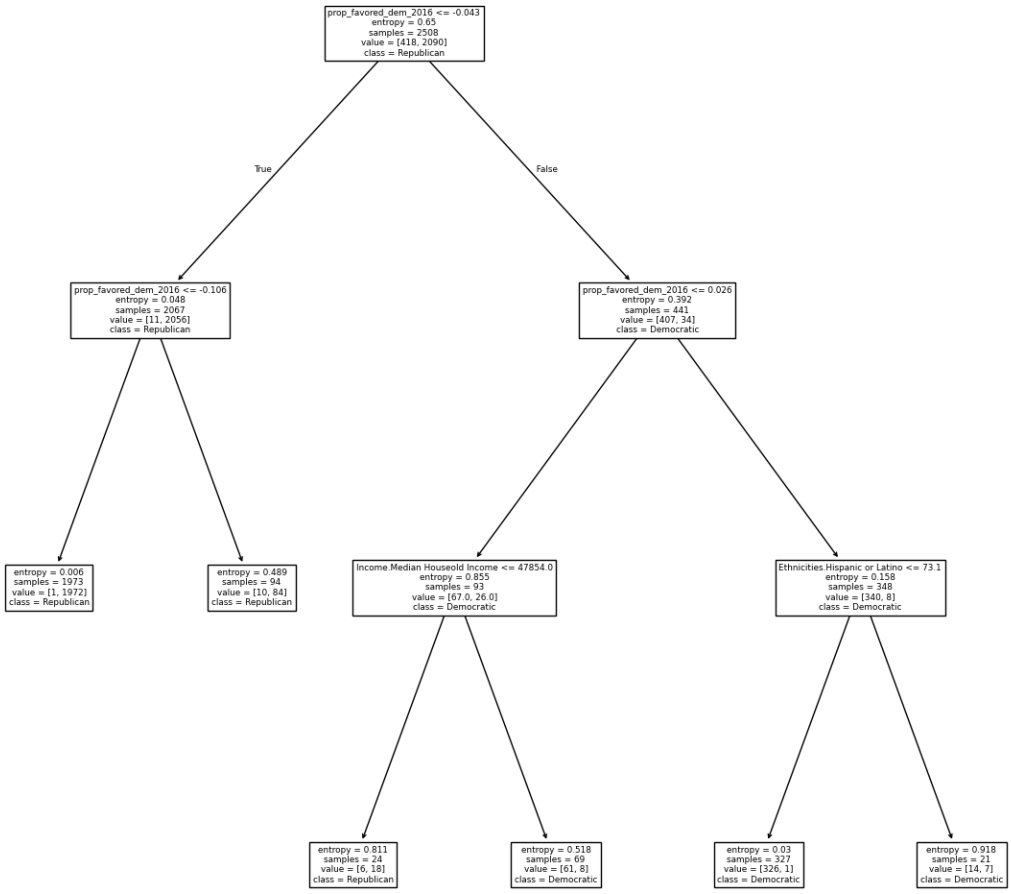


Figure 8: Pruned Decision Tree Using Past Election and Demographic Data

References

[1] D Irvani et al. "A discrete choice model of voters' decision in the general election 2019 using LASSO regression". In: *IOP Conference Series: Materials Science and Engineering* (2021). URL: https://iopscience.iop.org/article/10.1088/1757-899X/1115/1/012061/pdf.

[2] Marzuki et al. "Implementation of decision tree using C5.0 algorithm in preference and electability survey results on regional head election in Aceh". In: *Journal of Physics: Conference Series* (2021). DOI: 10.1088/1742-6596/1882/1/012132.

[3] Schmitz et al. "Could the outcome of the 2016 US elections have been predicted from past voting patterns?" In: *International Cartographic Association* (2017). URL: https://ica-proc.copernicus.org/articles/1/100/2018/ica-proc-1-100-2018.pdf.

[4] cinyc. "Alaska Presidential Results by County Equivalent, 1960-2020". In: *RRH Elections* (2021). URL: https://rrhelections.com/index.php/2021/04/13/alaska-presidential-results-by-county-equivalent-1960-2020/9/.

[5] MIT Election Data and Science Lab. "County Presidential Election Returns 2000-2020". In: *Harvard Dataverse* (2018). DOI: 10.7910/DVN/VOQCHQ.

[6] Saif Gazali and V. Pattabiraman. "Forecasting Election Data Using Regression Models and Sentimental Analysis". In: *Advances in Smart Grid Technology* (2021). Ed. by Ning Zhou and S. Hemamalini, pp. 501–509. URL: https://link.springer.com/chapter/10.1007/978-981-15-7241-8_37.

[7] Anthony Scime Gregg R. Murray. "Microtargeting and Electorate Segmentation: Data Mining the American National Election Studies". In: *Journal of Political Marketing* 9 (2010). DOI: 10.1080/15377857.2010.497732.

[8] Santiago Olivella Jacob M. Montgomery. "Tree-Based Models for Political Science Data". In: *American Journal of Political Science* 62 (2018). DOI: 10.1111/ajps.12361.

[9] Chitra Jegan. "Classification Of Diabetes Disease Using Support Vector Machine". In: *ResearchGate* (2013). URL: https://www.researchgate.net/publication/320395340_Classification_Of_Diabetes_Disease_Using_Support_Vector_Machine.

[10] Sanatan Mishra and John Lee. "Using Machine Learning Techniques to Predict United States House of Representatives Elections". In: *Journal of Student Research* 12 (2023). URL: https://www.jsr.org/hs/index.php/path/article/view/4697/2168.

[11] Bo Guan Ryan Whitcomb Joung Min Choi. "County Demographics CSV File". In: *CORGIS Dataset Project* (2021). URL: https://corgis-edu.github.io/corgis/csv/county_demographics/.

[12] Amit Choudhary Savita Ahlawat. "Hybrid CNN-SVM Classifier for Handwritten Digit Recognition". In: *Procedia Computer Science* 167 (2020). DOI: 10.1016/j.procs.2020.03.309.

[13] Simplemaps. "United States Counties Database". In: *simplemaps* (2024). URL: https://simplemaps.com/data/us-counties.

[14] Pankaj Sinha et al. "Forecasting United States Presidential election 2016 using multiple regression models". In: *Munich Personal RePEc Archive* (2016). URL: https://mpra.ub.uni-muenchen.de/74641/1/MPRA_paper_74641.pdf.

[15] Pankaj Sinha et al. "Prediction for the 2020 United States Presidential Election using Linear Regression Model". In: *Munich Personal RePEc Archive* (2020). URL: https://mpra.ub.uni-muenchen.de/103890/1/MPRA_paper_103890.pdf.

[16] Pankaj Sinha et al. "Prediction for the 2020 United States Presidential Election using Machine Learning Algorithm: Lasso Regression". In: *Munich Personal RePEc Archive* (2020). URL: https://mpra.ub.uni-muenchen.de/103889/1/MPRA_paper_103889.pdf.